

Supplier Corrosion - Inspection & Rework

Industrial Manufacturing | Incoming Quality / Supplier Defect Handling

October 5, 2026 | Anonymized real-world case

Confidentiality

This case study is anonymized. Company names, supplier names, internal identifiers, NCR numbers, specifications, part numbers and other confidential information have been omitted.

Problem	Supplier components arrived with corrosion in multiple areas of the ring.
My role	Physical quality assessment, OK/NOK judgement, corrosion removal/rework, rework decisions within my task, and post-rework verification.
Outcome	Recoverable components were reworked and verified. Some components could not be recovered because corrosion had penetrated deeply into the metal.
Formal QE ownership	No. I performed assigned inspection/rework activities and did not own supplier disposition, quality-system decisions or final release authority.

1. Background

On October 5, 2026, during incoming quality activities, I worked with supplier components that had arrived with visible corrosion. The affected parts included rings used for connection with a shaft.

Corrosion was present in multiple locations: on the sides, underneath the component, on the outer and inner surfaces, including inner side areas, and around threads.

2. Initial Quality Assessment

- I reviewed each component for the extent and location of corrosion.
- I determined whether the observed condition could be addressed through the assigned rework process.
- I made the practical OK/NOK assessment within my responsibility.
- The parts had arrived from the supplier with the observed corrosion; the defect was identified at incoming quality.

3. Rework Approach

For components that could be recovered, I carried out the required rework to remove corrosion. This included cleaning affected areas and working around and inside threaded areas where necessary.

I reviewed the condition during the process and continued rework where further corrosion remained rather than stopping after a single cleaning pass.

4. Verification After Rework

- After rework, I inspected the component step by step.
- I checked that visible corrosion had been completely removed, including difficult areas.
- I used visual inspection, including a flashlight, to check internal or less accessible surfaces.
- I observed the component over time to confirm that corrosion did not return.
- Where the condition remained unacceptable, the component was not treated as successfully reworked.

Verification principle: A component was considered successfully reworked only when the corrosion had been removed and did not return during the relevant observation period.

5. Non-Recoverable Components

Not every component could be recovered. At the time of this case, two parts remained completely NOK because the corrosion had penetrated deeply enough into the metal that the component had lost its intended usability.

This demonstrates the practical limit of rework: surface corrosion can be removed, but rework cannot restore a component whose material condition has already been seriously compromised.

6. My Contribution

- Performed incoming quality inspection of supplier components affected by corrosion.
- Assessed the location and severity of corrosion.
- Determined practical OK/NOK status within my assigned responsibility.
- Performed corrosion-removal rework on recoverable components.
- Adjusted the rework approach based on the condition observed during the work.
- Verified the completed rework and checked for recurrence.
- Identified cases where rework was no longer sufficient because the component was too deeply damaged.

7. Quality Capabilities Demonstrated

Capability	Evidence from the case
Defect identification	Recognized and assessed corrosion across multiple component locations.
OK/NOK judgement	Determined whether components were recoverable or remained unacceptable.
Rework execution	Performed practical corrosion removal, including work around difficult/threaded areas.
Verification	Checked the completed rework and whether corrosion returned.
Disposition awareness	Recognized when rework could no longer restore a component.

8. Connection to Quality Engineering Development

This case represents the inspection and execution side of Quality rather than formal Quality Engineering ownership. It complements my fluid-related corrosion investigation case, where the focus was hypothesis development, practical testing and evidence-based problem solving.

Portfolio classification

Real-world experience case | Anonymized / non-confidential | Formal QE ownership: No | Primary skills: Incoming Quality, Visual Inspection, OK/NOK Assessment, Rework, Verification, Defect Disposition Awareness